A DETAILED STATISTICAL RESULTS

Configuration-level descriptive statistics report sample size, median, mean, sample standard deviation, minimum, maximum, and sample variance. Standard deviations and variances use $n - 1$ denominators. Diagnostic tables report unadjusted Shapiro–Wilk and Levene results; comparison tables report test statistics, effect sizes, and raw and BH-adjusted p-values. All observations, including IQR-flagged values, are retained.

Table 9: Descriptive statistics: accuracy. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	66	66	0	66	66	0
Gemma 2 9B	MMLU	Q4_K_M	20	65	65	0	65	65	0
Gemma 2 9B	MMLU-Pro	FP16	20	49	49	0	49	49	0
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	49	49	0	49	49	0
Gemma 2 9B	RACE-H	FP16	20	91	91	0	91	91	0
Gemma 2 9B	RACE-H	Q4_K_M	20	93	93	0	93	93	0
Llama 3.1 8B	MMLU	FP16	20	52	52	0	52	52	0
Llama 3.1 8B	MMLU	Q4_K_M	20	57	57	7.29e-15	57	57	5.31e-29
Llama 3.1 8B	MMLU-Pro	FP16	20	43	43	0	43	43	0
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	48	48	0	48	48	0
Llama 3.1 8B	RACE-H	FP16	20	86	86	0	86	86	0
Llama 3.1 8B	RACE-H	Q4_K_M	20	86	86	0	86	86	0
Qwen 2.5 7B	MMLU	FP16	20	66	66	0	66	66	0
Qwen 2.5 7B	MMLU	Q4_K_M	20	61	61	0	61	61	0
Qwen 2.5 7B	MMLU-Pro	FP16	20	52	52	0	52	52	0
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	53	53	0	53	53	0
Qwen 2.5 7B	RACE-H	FP16	20	89	89	0	89	89	0
Qwen 2.5 7B	RACE-H	Q4_K_M	20	90	90	0	90	90	0

Table 10: Descriptive statistics: average power w. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	264.17	265.591	9.257	249.161	279.175	85.683
Gemma 2 9B	MMLU	Q4_K_M	20	244.633	243.801	7.322	228.981	258.075	53.61
Gemma 2 9B	MMLU-Pro	FP16	20	320.984	321.062	0.611	319.764	322.255	0.373
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	338.452	338.755	1.594	336.011	341.93	2.54
Gemma 2 9B	RACE-H	FP16	20	306.286	292.86	56.577	54.403	317.162	3200.921
Gemma 2 9B	RACE-H	Q4_K_M	20	289.121	290.136	8.613	274.765	305.584	74.181
Llama 3.1 8B	MMLU	FP16	20	257.315	256.512	6.616	246.475	272.988	43.766
Llama 3.1 8B	MMLU	Q4_K_M	20	237.101	236.25	8.577	220.425	253.447	73.564
Llama 3.1 8B	MMLU-Pro	FP16	20	314.231	314.401	1.158	311.322	316.077	1.341
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	354.613	354.466	1.852	351.059	357.195	3.431
Llama 3.1 8B	RACE-H	FP16	20	284.248	283.464	7.519	272.257	302.219	56.537
Llama 3.1 8B	RACE-H	Q4_K_M	20	280.48	281.078	11.222	258.574	298.002	125.928
Qwen 2.5 7B	MMLU	FP16	20	249.561	250.538	7.312	238.881	263.216	53.464
Qwen 2.5 7B	MMLU	Q4_K_M	20	223.235	223.135	9.679	205.888	238.811	93.682
Qwen 2.5 7B	MMLU-Pro	FP16	20	331.239	331.08	0.904	329.338	332.561	0.818
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	353.85	354.017	1.287	351.5	356.612	1.655
Qwen 2.5 7B	RACE-H	FP16	20	289.723	287.478	11.205	249.143	298.233	125.56
Qwen 2.5 7B	RACE-H	Q4_K_M	20	277.135	275.911	7.75	262.044	289.096	60.059

Table 11: Descriptive statistics: cpu temp end c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	62	62.15	1.843	59	65	3.397
Gemma 2 9B	MMLU	Q4_K_M	20	62	61.85	1.755	58	65	3.082
Gemma 2 9B	MMLU-Pro	FP16	20	70	70.2	1.642	67	75	2.695
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	70	69.75	1.943	67	76	3.776
Gemma 2 9B	RACE-H	FP16	20	63	63.55	2.282	61	68	5.208
Gemma 2 9B	RACE-H	Q4_K_M	20	62	62.2	2.016	59	69	4.063
Llama 3.1 8B	MMLU	FP16	20	62	62.6	1.465	60	65	2.147
Llama 3.1 8B	MMLU	Q4_K_M	20	62	62.3	1.809	60	67	3.274
Llama 3.1 8B	MMLU-Pro	FP16	20	69.5	69.9	2.469	67	75	6.095
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	70	70.35	1.089	68	73	1.187
Llama 3.1 8B	RACE-H	FP16	20	62	62.1	1.832	59	67	3.358
Llama 3.1 8B	RACE-H	Q4_K_M	20	62	61.9	1.165	61	66	1.358
Qwen 2.5 7B	MMLU	FP16	20	62	62.35	2.323	58	67	5.397
Qwen 2.5 7B	MMLU	Q4_K_M	20	63	63.35	2.641	59	72	6.976
Qwen 2.5 7B	MMLU-Pro	FP16	20	71	70.8	1.989	68	74	3.958
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	71	70.95	1.932	68	75	3.734
Qwen 2.5 7B	RACE-H	FP16	20	62	62.7	1.75	60	66	3.063
Qwen 2.5 7B	RACE-H	Q4_K_M	20	62	62.85	2.758	58	68	7.608

Table 12: Descriptive statistics: cpu temp max c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	70	69.7	2.658	63	74	7.063
Gemma 2 9B	MMLU	Q4_K_M	20	67	67.8	1.989	65	72	3.958
Gemma 2 9B	MMLU-Pro	FP16	20	74	74.6	3.102	70	84	9.621
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	73.5	74.3	3.114	71	83	9.695
Gemma 2 9B	RACE-H	FP16	20	67	67.3	2.386	64	73	5.695
Gemma 2 9B	RACE-H	Q4_K_M	20	68.5	68.95	3.692	63	77	13.629
Llama 3.1 8B	MMLU	FP16	20	68	68.45	2.328	63	72	5.418
Llama 3.1 8B	MMLU	Q4_K_M	20	67.5	69.05	3.379	65	78	11.418
Llama 3.1 8B	MMLU-Pro	FP16	20	74	74.3	2.203	71	81	4.853
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	73	73.2	1.824	71	78	3.326
Llama 3.1 8B	RACE-H	FP16	20	67.5	67.4	2.303	63	71	5.305
Llama 3.1 8B	RACE-H	Q4_K_M	20	67.5	67.85	2.3	64	71	5.292
Qwen 2.5 7B	MMLU	FP16	20	67	68.2	3.302	65	79	10.905
Qwen 2.5 7B	MMLU	Q4_K_M	20	70	69.75	3.768	65	78	14.197
Qwen 2.5 7B	MMLU-Pro	FP16	20	74	75.05	2.35	73	81	5.524
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	73	74.1	2.1	71	79	4.411
Qwen 2.5 7B	RACE-H	FP16	20	67.5	68.25	2.673	64	73	7.145
Qwen 2.5 7B	RACE-H	Q4_K_M	20	67.5	68.1	2.15	65	73	4.621

Table 13: Descriptive statistics: cpu temp mean c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	62.55	62.455	1.122	60.526	64.27	1.258
Gemma 2 9B	MMLU	Q4_K_M	20	62.573	62.809	0.921	61.669	64.558	0.848
Gemma 2 9B	MMLU-Pro	FP16	20	68.027	68.078	0.527	66.7	69.204	0.277
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	67.162	67.339	0.94	66.019	69.029	0.883
Gemma 2 9B	RACE-H	FP16	20	62.342	62.491	0.99	61.291	64.648	0.98
Gemma 2 9B	RACE-H	Q4_K_M	20	62.795	62.926	1.528	60.672	68.566	2.336
Llama 3.1 8B	MMLU	FP16	20	62.382	62.295	1.083	60.871	64.42	1.174
Llama 3.1 8B	MMLU	Q4_K_M	20	62.816	63.126	1.315	61.29	66.478	1.73
Llama 3.1 8B	MMLU-Pro	FP16	20	68.591	68.552	0.769	67.133	70.127	0.591
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	67.478	67.559	0.819	66.283	68.837	0.67
Llama 3.1 8B	RACE-H	FP16	20	62.222	62.313	0.731	61.092	63.673	0.534
Llama 3.1 8B	RACE-H	Q4_K_M	20	62.171	62.537	0.934	61.076	63.956	0.872
Qwen 2.5 7B	MMLU	FP16	20	62.559	62.701	1.669	60.684	68.347	2.787
Qwen 2.5 7B	MMLU	Q4_K_M	20	63.891	63.632	1.245	61.968	65.841	1.551
Qwen 2.5 7B	MMLU-Pro	FP16	20	69.285	69.194	0.614	67.595	70.029	0.377
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	68.393	68.364	0.83	66.042	69.591	0.689
Qwen 2.5 7B	RACE-H	FP16	20	61.871	62.165	1.17	59.683	64.787	1.369
Qwen 2.5 7B	RACE-H	Q4_K_M	20	62.405	62.82	1.443	61.399	67.847	2.084

Table 14: Descriptive statistics: cpu temp start c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	65	64.55	3.154	59	69	9.945
Gemma 2 9B	MMLU	Q4_K_M	20	63	63.6	2.21	61	69	4.884
Gemma 2 9B	MMLU-Pro	FP16	20	63	63.75	2.337	61	70	5.461
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	65	64.7	2.003	62	69	4.011
Gemma 2 9B	RACE-H	FP16	20	64.5	64.9	2.732	61	69	7.463
Gemma 2 9B	RACE-H	Q4_K_M	20	64.5	64.5	1.67	62	68	2.789
Llama 3.1 8B	MMLU	FP16	20	65	63.5	4.628	46	69	21.421
Llama 3.1 8B	MMLU	Q4_K_M	20	63	63.55	1.731	59	66	2.997
Llama 3.1 8B	MMLU-Pro	FP16	20	65	64.9	2.469	60	70	6.095
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	63	63.75	1.585	61	66	2.513
Llama 3.1 8B	RACE-H	FP16	20	64	64.45	1.791	62	69	3.208
Llama 3.1 8B	RACE-H	Q4_K_M	20	65	64.5	1.762	62	67	3.105
Qwen 2.5 7B	MMLU	FP16	20	65	64.55	2.259	61	69	5.103
Qwen 2.5 7B	MMLU	Q4_K_M	20	65	64.9	2.292	62	70	5.253
Qwen 2.5 7B	MMLU-Pro	FP16	20	65	64.1	4.887	45	70	23.884
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	64	63.9	2.198	58	67	4.832
Qwen 2.5 7B	RACE-H	FP16	20	65	64.5	2.236	60	67	5
Qwen 2.5 7B	RACE-H	Q4_K_M	20	65	64.65	2.134	62	72	4.555

Table 15: Descriptive statistics: cpu utilization pct. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	7.024	7.024	0.043	6.925	7.092	0.002
Gemma 2 9B	MMLU	Q4_K_M	20	7.118	7.133	0.065	7.045	7.299	0.004
Gemma 2 9B	MMLU-Pro	FP16	20	6.823	6.823	0.013	6.799	6.85	0.000161
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	6.85	6.85	0.014	6.821	6.872	0.000208
Gemma 2 9B	RACE-H	FP16	20	7.047	7.074	0.13	6.963	7.587	0.017
Gemma 2 9B	RACE-H	Q4_K_M	20	7.101	7.097	0.046	6.976	7.188	0.002
Llama 3.1 8B	MMLU	FP16	20	7.005	7.014	0.047	6.944	7.115	0.002
Llama 3.1 8B	MMLU	Q4_K_M	20	7.178	7.186	0.053	7.113	7.301	0.003
Llama 3.1 8B	MMLU-Pro	FP16	20	6.817	6.819	0.013	6.792	6.854	0.000157
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	6.862	6.862	0.014	6.841	6.895	0.000205
Llama 3.1 8B	RACE-H	FP16	20	7.073	7.083	0.048	7.013	7.212	0.002
Llama 3.1 8B	RACE-H	Q4_K_M	20	7.164	7.137	0.147	6.575	7.278	0.022
Qwen 2.5 7B	MMLU	FP16	20	7.063	7.073	0.047	7.009	7.145	0.002
Qwen 2.5 7B	MMLU	Q4_K_M	20	7.178	7.179	0.057	7.096	7.272	0.003
Qwen 2.5 7B	MMLU-Pro	FP16	20	6.819	6.821	0.015	6.792	6.847	0.000234
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	6.86	6.858	0.015	6.832	6.888	0.000238
Qwen 2.5 7B	RACE-H	FP16	20	7.09	7.067	0.121	6.598	7.168	0.015
Qwen 2.5 7B	RACE-H	Q4_K_M	20	7.124	7.14	0.066	7.037	7.306	0.004

Table 16: Descriptive statistics: gpu temp end c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	45.5	45.55	1.191	44	47	1.418
Gemma 2 9B	MMLU	Q4_K_M	20	45	44.6	1.046	43	46	1.095
Gemma 2 9B	MMLU-Pro	FP16	20	50	50	0	50	50	0
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	50	50.05	0.394	49	51	0.155
Gemma 2 9B	RACE-H	FP16	20	47	46.6	1.569	41	48	2.463
Gemma 2 9B	RACE-H	Q4_K_M	20	47	46.55	1.146	44	49	1.313
Llama 3.1 8B	MMLU	FP16	20	45.5	45.45	1.317	43	47	1.734
Llama 3.1 8B	MMLU	Q4_K_M	20	45	44.65	1.089	43	46	1.187
Llama 3.1 8B	MMLU-Pro	FP16	20	50	50	0	50	50	0
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	50	50.4	0.503	50	51	0.253
Llama 3.1 8B	RACE-H	FP16	20	46	46	0.973	44	48	0.947
Llama 3.1 8B	RACE-H	Q4_K_M	20	46	45.9	1.119	44	47	1.253
Qwen 2.5 7B	MMLU	FP16	20	45	45.3	1.218	43	48	1.484
Qwen 2.5 7B	MMLU	Q4_K_M	20	45	45.2	0.894	43	46	0.8
Qwen 2.5 7B	MMLU-Pro	FP16	20	50	50.25	0.444	50	51	0.197
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	50	50.3	0.47	50	51	0.221
Qwen 2.5 7B	RACE-H	FP16	20	46	45.9	1.294	44	48	1.674
Qwen 2.5 7B	RACE-H	Q4_K_M	20	46	45.6	1.501	43	49	2.253

Table 17: Descriptive statistics: gpu temp max c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	59	58.65	1.424	56	61	2.029
Gemma 2 9B	MMLU	Q4_K_M	20	59	58.75	1.682	56	62	2.829
Gemma 2 9B	MMLU-Pro	FP16	20	64.5	64.55	0.759	63	66	0.576
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	65	65.25	0.967	63	67	0.934
Gemma 2 9B	RACE-H	FP16	20	62	61.55	4.478	43	64	20.05
Gemma 2 9B	RACE-H	Q4_K_M	20	63.5	63.1	1.334	61	66	1.779
Llama 3.1 8B	MMLU	FP16	20	57	57.15	1.694	55	60	2.871
Llama 3.1 8B	MMLU	Q4_K_M	20	58	57.75	1.41	55	60	1.987
Llama 3.1 8B	MMLU-Pro	FP16	20	63	63.35	1.531	60	66	2.345
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	65	64.7	1.031	63	66	1.063
Llama 3.1 8B	RACE-H	FP16	20	61	60.85	1.04	59	63	1.082
Llama 3.1 8B	RACE-H	Q4_K_M	20	62	62.1	1.252	60	64	1.568
Qwen 2.5 7B	MMLU	FP16	20	58	57.95	1.791	55	61	3.208
Qwen 2.5 7B	MMLU	Q4_K_M	20	57	57.45	1.432	55	61	2.05
Qwen 2.5 7B	MMLU-Pro	FP16	20	65	64.55	0.759	63	66	0.576
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	64.5	64.45	1.099	63	66	1.208
Qwen 2.5 7B	RACE-H	FP16	20	61	61	1.522	58	63	2.316
Qwen 2.5 7B	RACE-H	Q4_K_M	20	61	61.35	1.496	59	65	2.239

Table 18: Descriptive statistics: gpu temp mean c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	49.885	49.955	1.14	48.158	51.552	1.301
Gemma 2 9B	MMLU	Q4_K_M	20	48.919	48.722	1.196	46.487	50.519	1.431
Gemma 2 9B	MMLU-Pro	FP16	20	57.813	57.797	0.291	57.309	58.526	0.085
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	59.601	59.484	0.517	58.435	60.219	0.267
Gemma 2 9B	RACE-H	FP16	20	52.981	52.501	2.855	41.105	54.631	8.151
Gemma 2 9B	RACE-H	Q4_K_M	20	53.569	53.465	1.457	50.677	57.308	2.122
Llama 3.1 8B	MMLU	FP16	20	49.69	49.333	1.286	47.067	50.899	1.654
Llama 3.1 8B	MMLU	Q4_K_M	20	48.486	48.481	1.087	46.848	50.021	1.181
Llama 3.1 8B	MMLU-Pro	FP16	20	57.184	57.149	0.388	56.581	57.618	0.15
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	60.349	60.349	0.482	59.341	60.992	0.232
Llama 3.1 8B	RACE-H	FP16	20	51.321	51.31	1.044	49.408	52.854	1.091
Llama 3.1 8B	RACE-H	Q4_K_M	20	52.604	52.056	1.387	48.739	53.722	1.924
Qwen 2.5 7B	MMLU	FP16	20	49.186	49.093	1.41	46.491	52.733	1.989
Qwen 2.5 7B	MMLU	Q4_K_M	20	48.361	48.198	0.788	46.413	49.119	0.622
Qwen 2.5 7B	MMLU-Pro	FP16	20	58.593	58.561	0.166	58.288	58.91	0.027
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	60.793	60.696	0.373	59.788	61.174	0.139
Qwen 2.5 7B	RACE-H	FP16	20	51.395	51.363	1.293	49.256	53.458	1.671
Qwen 2.5 7B	RACE-H	Q4_K_M	20	51.744	51.588	1.627	48.873	56.115	2.648

Table 19: Descriptive statistics: gpu temp start c. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	42	42.55	1.146	41	44	1.313
Gemma 2 9B	MMLU	Q4_K_M	20	42	42.15	1.182	40	44	1.397
Gemma 2 9B	MMLU-Pro	FP16	20	43	43.05	1.761	41	49	3.103
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	43	42.6	1.392	40	44	1.937
Gemma 2 9B	RACE-H	FP16	20	43	42.9	0.912	41	44	0.832
Gemma 2 9B	RACE-H	Q4_K_M	20	42.5	42.8	1.824	40	49	3.326
Llama 3.1 8B	MMLU	FP16	20	42.5	41.9	1.21	40	43	1.463
Llama 3.1 8B	MMLU	Q4_K_M	20	42	42	1.076	40	44	1.158
Llama 3.1 8B	MMLU-Pro	FP16	20	41.5	42.05	1.905	40	48	3.629
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	41.5	41.65	1.309	39	43	1.713
Llama 3.1 8B	RACE-H	FP16	20	42	41.85	0.933	40	43	0.871
Llama 3.1 8B	RACE-H	Q4_K_M	20	42.5	42.15	1.182	40	44	1.397
Qwen 2.5 7B	MMLU	FP16	20	43	42.85	1.981	40	49	3.924
Qwen 2.5 7B	MMLU	Q4_K_M	20	43	42.75	0.91	41	44	0.829
Qwen 2.5 7B	MMLU-Pro	FP16	20	43	42.5	1.235	41	44	1.526
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	42	41.55	1.234	40	43	1.524
Qwen 2.5 7B	RACE-H	FP16	20	42	42.2	1.281	40	44	1.642
Qwen 2.5 7B	RACE-H	Q4_K_M	20	42	41.95	1.932	40	48	3.734

Table 20: Descriptive statistics: gpu utilization pct. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	60.502	60.543	0.399	59.418	61.123	0.159
Gemma 2 9B	MMLU	Q4_K_M	20	41.532	41.479	0.315	40.84	42.138	0.099
Gemma 2 9B	MMLU-Pro	FP16	20	92.571	92.571	0.025	92.536	92.62	0.000637
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	83.373	83.381	0.06	83.277	83.492	0.004
Gemma 2 9B	RACE-H	FP16	20	65.561	62.497	14.019	2.947	66.037	196.529
Gemma 2 9B	RACE-H	Q4_K_M	20	52.677	52.763	0.318	52.293	53.477	0.101
Llama 3.1 8B	MMLU	FP16	20	62.535	62.603	0.202	62.241	63.034	0.041
Llama 3.1 8B	MMLU	Q4_K_M	20	37.254	37.16	0.237	36.721	37.428	0.056
Llama 3.1 8B	MMLU-Pro	FP16	20	94.718	94.713	0.028	94.617	94.744	0.000806
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	87.203	87.198	0.055	87.086	87.295	0.003
Llama 3.1 8B	RACE-H	FP16	20	58.3	58.301	0.3	57.952	58.888	0.09
Llama 3.1 8B	RACE-H	Q4_K_M	20	46.009	45.431	1.625	38.691	46.222	2.642
Qwen 2.5 7B	MMLU	FP16	20	50.387	50.345	0.309	49.977	50.867	0.095
Qwen 2.5 7B	MMLU	Q4_K_M	20	31.996	31.912	0.238	31.492	32.317	0.056
Qwen 2.5 7B	MMLU-Pro	FP16	20	94.409	94.404	0.02	94.376	94.433	0.000381
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	86.669	86.667	0.042	86.564	86.733	0.002
Qwen 2.5 7B	RACE-H	FP16	20	56.485	56.207	1.682	49.112	56.95	2.831
Qwen 2.5 7B	RACE-H	Q4_K_M	20	44.481	44.629	0.34	43.837	45.329	0.116

Table 21: Descriptive statistics: inference time s. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	18	18.05	0.224	18	19	0.05
Gemma 2 9B	MMLU	Q4_K_M	20	11	10.7	0.47	10	11	0.221
Gemma 2 9B	MMLU-Pro	FP16	20	508	507.95	0.224	507	508	0.05
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	217	216.6	0.503	216	217	0.253
Gemma 2 9B	RACE-H	FP16	20	22	22.15	0.366	22	23	0.134
Gemma 2 9B	RACE-H	Q4_K_M	20	15	15.2	0.41	15	16	0.168
Llama 3.1 8B	MMLU	FP16	20	19	19.1	0.308	19	20	0.095
Llama 3.1 8B	MMLU	Q4_K_M	20	9	9.1	0.308	9	10	0.095
Llama 3.1 8B	MMLU-Pro	FP16	20	617	616.9	0.308	616	617	0.095
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	223	222.8	0.41	222	223	0.168
Llama 3.1 8B	RACE-H	FP16	20	16	16.1	0.308	16	17	0.095
Llama 3.1 8B	RACE-H	Q4_K_M	20	11	11.15	0.366	11	12	0.134
Qwen 2.5 7B	MMLU	FP16	20	13	12.65	0.489	12	13	0.239
Qwen 2.5 7B	MMLU	Q4_K_M	20	8	7.9	0.308	7	8	0.095
Qwen 2.5 7B	MMLU-Pro	FP16	20	758	757.6	0.503	757	758	0.253
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	316	316.1	0.308	316	317	0.095
Qwen 2.5 7B	RACE-H	FP16	20	15.5	15.5	0.513	15	16	0.263
Qwen 2.5 7B	RACE-H	Q4_K_M	20	11	11.05	0.224	11	12	0.05

Table 22: Descriptive statistics: output tokens. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	595	595	0	595	595	0
Gemma 2 9B	MMLU	Q4_K_M	20	591	591	0	591	591	0
Gemma 2 9B	MMLU-Pro	FP16	20	21752	21752	0	21752	21752	0
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	20933	20933	0	20933	20933	0
Gemma 2 9B	RACE-H	FP16	20	600	600	0	600	600	0
Gemma 2 9B	RACE-H	Q4_K_M	20	600	600	0	600	600	0
Llama 3.1 8B	MMLU	FP16	20	833	833	0	833	833	0
Llama 3.1 8B	MMLU	Q4_K_M	20	643	643	0	643	643	0
Llama 3.1 8B	MMLU-Pro	FP16	20	34142	34142	0	34142	34142	0
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	30000	30000	0	30000	30000	0
Llama 3.1 8B	RACE-H	FP16	20	500	500	0	500	500	0
Llama 3.1 8B	RACE-H	Q4_K_M	20	500	500	0	500	500	0
Qwen 2.5 7B	MMLU	FP16	20	500	500	0	500	500	0
Qwen 2.5 7B	MMLU	Q4_K_M	20	500	500	0	500	500	0
Qwen 2.5 7B	MMLU-Pro	FP16	20	44331	44331	0	44331	44331	0
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	44559	44559	0	44559	44559	0
Qwen 2.5 7B	RACE-H	FP16	20	500	500	0	500	500	0
Qwen 2.5 7B	RACE-H	Q4_K_M	20	500	500	0	500	500	0

Table 23: Descriptive statistics: peak gpu memory mb. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	21617.188	21617.088	0.447	21615.188	21617.188	0.2
Gemma 2 9B	MMLU	Q4_K_M	20	9521.188	9520.987	0.616	9519.188	9521.188	0.379
Gemma 2 9B	MMLU-Pro	FP16	20	21617.188	21617.188	0	21617.188	21617.188	0
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	9521.188	9521.188	0	9521.188	9521.188	0
Gemma 2 9B	RACE-H	FP16	20	21619.188	21618.287	4.025	21601.188	21619.188	16.2
Gemma 2 9B	RACE-H	Q4_K_M	20	9523.188	9523.188	0	9523.188	9523.188	0
Llama 3.1 8B	MMLU	FP16	20	16691.188	16690.588	0.94	16689.188	16691.188	0.884
Llama 3.1 8B	MMLU	Q4_K_M	20	6825.188	6825.188	0	6825.188	6825.188	0
Llama 3.1 8B	MMLU-Pro	FP16	20	16689.188	16689.188	0	16689.188	16689.188	0
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	6823.188	6823.188	0	6823.188	6823.188	0
Llama 3.1 8B	RACE-H	FP16	20	16697.188	16697.188	0	16697.188	16697.188	0
Llama 3.1 8B	RACE-H	Q4_K_M	20	6831.188	6831.188	0	6831.188	6831.188	0
Qwen 2.5 7B	MMLU	FP16	20	15221.188	15220.888	0.733	15219.188	15221.188	0.537
Qwen 2.5 7B	MMLU	Q4_K_M	20	5947.188	5946.988	0.616	5945.188	5947.188	0.379
Qwen 2.5 7B	MMLU-Pro	FP16	20	15219.188	15219.188	0	15219.188	15219.188	0
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	5945.188	5945.188	0	5945.188	5945.188	0
Qwen 2.5 7B	RACE-H	FP16	20	15223.188	15223.188	0	15223.188	15223.188	0
Qwen 2.5 7B	RACE-H	Q4_K_M	20	5951.188	5951.188	0	5951.188	5951.188	0

Table 24: Descriptive statistics: peak ram mb. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	4491.457	4474.383	77.529	4297.605	4578.738	6010.714
Gemma 2 9B	MMLU	Q4_K_M	20	3425.223	3421.112	67.865	3276.93	3523.531	4605.712
Gemma 2 9B	MMLU-Pro	FP16	20	4522.732	4493.361	81.523	4334.789	4605.43	6646.063
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	3442.777	3446.955	77.459	3319.656	3554.242	5999.872
Gemma 2 9B	RACE-H	FP16	20	4379.441	4375.91	78.837	4198.57	4501.191	6215.347
Gemma 2 9B	RACE-H	Q4_K_M	20	3428.229	3402.89	97.443	3257.738	3559.688	9495.115
Llama 3.1 8B	MMLU	FP16	20	3646.703	3607.479	75.076	3448.457	3701.551	5636.473
Llama 3.1 8B	MMLU	Q4_K_M	20	2914.34	2900.686	80.064	2745.473	3011.324	6410.197
Llama 3.1 8B	MMLU-Pro	FP16	20	3640.637	3623.837	85.302	3469.258	3739.629	7276.457
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	2992.676	2944.844	81.097	2769.316	3013.676	6576.686
Llama 3.1 8B	RACE-H	FP16	20	3611.377	3597.17	84.689	3445.762	3729.035	7172.147
Llama 3.1 8B	RACE-H	Q4_K_M	20	2952.256	2909.556	81.911	2766.734	3027.539	6709.475
Qwen 2.5 7B	MMLU	FP16	20	3718.561	3682.448	79.053	3499.07	3775.414	6249.394
Qwen 2.5 7B	MMLU	Q4_K_M	20	2984.254	2940.475	88.578	2756.926	3044.828	7846.065
Qwen 2.5 7B	MMLU-Pro	FP16	20	3701.375	3655.769	85.465	3516.117	3740.875	7304.218
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	2922.734	2919.063	77.824	2788.984	3030.977	6056.621
Qwen 2.5 7B	RACE-H	FP16	20	3664.938	3622.492	90.519	3452.902	3753.48	8193.65
Qwen 2.5 7B	RACE-H	Q4_K_M	20	2947.781	2905.395	82.653	2736.727	3004.688	6831.464

Table 25: Descriptive statistics: run time s. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	21.616	21.611	0.073	21.459	21.771	0.005
Gemma 2 9B	MMLU	Q4_K_M	20	14.1	14.1	0.062	13.964	14.194	0.004
Gemma 2 9B	MMLU-Pro	FP16	20	511.124	511.119	0.088	510.928	511.268	0.008
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	219.91	219.912	0.096	219.725	220.063	0.009
Gemma 2 9B	RACE-H	FP16	20	25.569	25.578	0.06	25.487	25.681	0.004
Gemma 2 9B	RACE-H	Q4_K_M	20	18.426	18.43	0.058	18.339	18.513	0.003
Llama 3.1 8B	MMLU	FP16	20	22.629	22.635	0.078	22.491	22.82	0.006
Llama 3.1 8B	MMLU	Q4_K_M	20	12.531	12.533	0.065	12.438	12.696	0.004
Llama 3.1 8B	MMLU-Pro	FP16	20	620.228	620.264	0.172	620.11	620.92	0.03
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	226.029	226.041	0.095	225.924	226.347	0.009
Llama 3.1 8B	RACE-H	FP16	20	19.365	19.365	0.068	19.26	19.516	0.005
Llama 3.1 8B	RACE-H	Q4_K_M	20	14.556	14.71	0.674	14.469	17.552	0.454
Qwen 2.5 7B	MMLU	FP16	20	16.158	16.167	0.078	16.047	16.274	0.006
Qwen 2.5 7B	MMLU	Q4_K_M	20	11.323	11.336	0.081	11.206	11.523	0.007
Qwen 2.5 7B	MMLU-Pro	FP16	20	760.77	760.801	0.14	760.553	761.11	0.02
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	319.418	319.424	0.134	319.21	319.764	0.018
Qwen 2.5 7B	RACE-H	FP16	20	18.807	18.96	0.68	18.659	21.831	0.462
Qwen 2.5 7B	RACE-H	Q4_K_M	20	14.405	14.418	0.067	14.331	14.639	0.004

Table 26: Descriptive statistics: throughput tokens s. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	33.056	32.969	0.389	31.316	33.056	0.151
Gemma 2 9B	MMLU	Q4_K_M	20	53.727	55.339	2.526	53.727	59.1	6.381
Gemma 2 9B	MMLU-Pro	FP16	20	42.819	42.823	0.019	42.819	42.903	0.000357
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	96.465	96.644	0.224	96.465	96.912	0.05
Gemma 2 9B	RACE-H	FP16	20	27.273	27.095	0.434	26.087	27.273	0.189
Gemma 2 9B	RACE-H	Q4_K_M	20	40	39.5	1.026	37.5	40	1.053
Llama 3.1 8B	MMLU	FP16	20	43.842	43.623	0.675	41.65	43.842	0.455
Llama 3.1 8B	MMLU	Q4_K_M	20	71.444	70.73	2.199	64.3	71.444	4.836
Llama 3.1 8B	MMLU-Pro	FP16	20	55.335	55.344	0.028	55.335	55.425	0.000764
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	134.529	134.65	0.249	134.529	135.135	0.062
Llama 3.1 8B	RACE-H	FP16	20	31.25	31.066	0.566	29.412	31.25	0.32
Llama 3.1 8B	RACE-H	Q4_K_M	20	45.455	44.886	1.388	41.667	45.455	1.926
Qwen 2.5 7B	MMLU	FP16	20	38.462	39.583	1.568	38.462	41.667	2.46
Qwen 2.5 7B	MMLU	Q4_K_M	20	62.5	63.393	2.748	62.5	71.429	7.552
Qwen 2.5 7B	MMLU-Pro	FP16	20	58.484	58.515	0.039	58.484	58.561	0.002
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	141.009	140.965	0.137	140.565	141.009	0.019
Qwen 2.5 7B	RACE-H	FP16	20	32.292	32.292	1.069	31.25	33.333	1.142
Qwen 2.5 7B	RACE-H	Q4_K_M	20	45.455	45.265	0.847	41.667	45.455	0.717

Table 27: Descriptive statistics: token limit rate. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	0	0	0	0	0	0
Gemma 2 9B	MMLU	Q4_K_M	20	0	0	0	0	0	0
Gemma 2 9B	MMLU-Pro	FP16	20	0	0	0	0	0	0
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0	0	0	0	0	0
Gemma 2 9B	RACE-H	FP16	20	0	0	0	0	0	0
Gemma 2 9B	RACE-H	Q4_K_M	20	0	0	0	0	0	0
Llama 3.1 8B	MMLU	FP16	20	0.3	0.3	5.7e-17	0.3	0.3	3.24e-33
Llama 3.1 8B	MMLU	Q4_K_M	20	0.13	0.13	2.85e-17	0.13	0.13	8.11e-34
Llama 3.1 8B	MMLU-Pro	FP16	20	0.11	0.11	1.42e-17	0.11	0.11	2.03e-34
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.06	0.06	1.42e-17	0.06	0.06	2.03e-34
Llama 3.1 8B	RACE-H	FP16	20	0	0	0	0	0	0
Llama 3.1 8B	RACE-H	Q4_K_M	20	0	0	0	0	0	0
Qwen 2.5 7B	MMLU	FP16	20	0	0	0	0	0	0
Qwen 2.5 7B	MMLU	Q4_K_M	20	0	0	0	0	0	0
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.01	0.01	1.78e-18	0.01	0.01	3.17e-36
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.06	0.06	1.42e-17	0.06	0.06	2.03e-34
Qwen 2.5 7B	RACE-H	FP16	20	0	0	0	0	0	0
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0	0	0	0	0	0

Table 28: Descriptive statistics: total energy j. SD and variance use ddof=1.

Model	Benchmark	Precision	n	Median	Mean	SD	Min	Max	Variance
Gemma 2 9B	MMLU	FP16	20	5722.691	5739.637	198.195	5396.815	6077.823	39281.106
Gemma 2 9B	MMLU	Q4_K_M	20	3448.489	3437.64	102.063	3232.923	3642.589	10416.949
Gemma 2 9B	MMLU-Pro	FP16	20	164037.277	164100.739	313.421	163443.22	164739.058	98232.884
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	74417.791	74496.372	359.154	73912.243	75218.538	128991.633
Gemma 2 9B	RACE-H	FP16	20	7827.351	7490.394	1446.138	1393.284	8108.248	2091315.898
Gemma 2 9B	RACE-H	Q4_K_M	20	5316.866	5347.205	155.539	5086.684	5649.08	24192.279
Llama 3.1 8B	MMLU	FP16	20	5799.507	5806.049	149.406	5588.126	6182.03	22322.287
Llama 3.1 8B	MMLU	Q4_K_M	20	2956.685	2960.564	98.552	2798.456	3160.38	9712.41
Llama 3.1 8B	MMLU-Pro	FP16	20	194958.113	195011.516	719.772	193116.348	196047.955	518071.28
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	80224.328	80123.76	409.577	79328.711	80700.799	167753.427
Llama 3.1 8B	RACE-H	FP16	20	5482.463	5489.345	149.175	5277.193	5846.896	22253.261
Llama 3.1 8B	RACE-H	Q4_K_M	20	4119.404	4131.25	176.391	3788.02	4564.951	31113.824
Qwen 2.5 7B	MMLU	FP16	20	4043.269	4050.505	118.947	3841.939	4267.975	14148.398
Qwen 2.5 7B	MMLU	Q4_K_M	20	2530.706	2529.102	102.526	2350.051	2688.419	10511.628
Qwen 2.5 7B	MMLU-Pro	FP16	20	251975.319	251885.768	703.557	250493.584	253081.487	494992.721
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	113017.001	113081.45	426.457	112255.464	114031.932	181865.813
Qwen 2.5 7B	RACE-H	FP16	20	5442.303	5444.672	117.302	5119.889	5614.637	13759.683
Qwen 2.5 7B	RACE-H	Q4_K_M	20	3985.864	3978.029	109.994	3790.669	4159.148	12098.727

Table 29: Shapiro–Wilk checks: average power w. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.961	0.556	not rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.966	0.665	not rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.941	0.255	not rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.939	0.233	not rejected
Llama 3.1 8B	RACE-H	FP16	20	0.945	0.297	not rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.954	0.433	not rejected
Qwen 2.5 7B	MMLU	FP16	20	0.941	0.255	not rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.958	0.506	not rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.974	0.837	not rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.975	0.846	not rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.746	0.00015	rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.951	0.388	not rejected
Gemma 2 9B	MMLU	FP16	20	0.949	0.351	not rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.962	0.585	not rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.954	0.434	not rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.975	0.858	not rejected
Gemma 2 9B	RACE-H	FP16	20	0.347	1.71e-08	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.979	0.919	not rejected

Table 30: Shapiro–Wilk checks: cpu utilization pct. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.964	0.63	not rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.935	0.196	not rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.925	0.125	not rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.95	0.364	not rejected
Llama 3.1 8B	RACE-H	FP16	20	0.921	0.103	not rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.657	1.22e-05	rejected
Qwen 2.5 7B	MMLU	FP16	20	0.92	0.098	not rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.936	0.205	not rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.968	0.72	not rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.975	0.851	not rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.621	4.85e-06	rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.904	0.048	rejected
Gemma 2 9B	MMLU	FP16	20	0.966	0.666	not rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.932	0.168	not rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.981	0.944	not rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.966	0.677	not rejected
Gemma 2 9B	RACE-H	FP16	20	0.59	2.34e-06	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.948	0.342	not rejected

Table 31: Shapiro–Wilk checks: gpu utilization pct. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.91	0.062	not rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.772	0.000344	rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.812	0.001	rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.967	0.691	not rejected
Llama 3.1 8B	RACE-H	FP16	20	0.873	0.013	rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.414	5.78e-08	rejected
Qwen 2.5 7B	MMLU	FP16	20	0.88	0.017	rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.792	0.000658	rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.904	0.05	rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.953	0.422	not rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.343	1.6e-08	rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.892	0.029	rejected
Gemma 2 9B	MMLU	FP16	20	0.912	0.069	not rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.943	0.279	not rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.955	0.457	not rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.954	0.427	not rejected
Gemma 2 9B	RACE-H	FP16	20	0.251	3.44e-09	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.757	0.00021	rejected

Table 32: Shapiro–Wilk checks: inference time s. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.351	1.85e-08	rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.351	1.85e-08	rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.351	1.85e-08	rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.495	2.89e-07	rejected
Llama 3.1 8B	RACE-H	FP16	20	0.351	1.85e-08	rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.433	8.32e-08	rejected
Qwen 2.5 7B	MMLU	FP16	20	0.608	3.54e-06	rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.351	1.85e-08	rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.626	5.6e-06	rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.351	1.85e-08	rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.641	8.1e-06	rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU	FP16	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.58	1.86e-06	rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.626	5.6e-06	rejected
Gemma 2 9B	RACE-H	FP16	20	0.433	8.32e-08	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.495	2.89e-07	rejected

Table 33: Shapiro–Wilk checks: peak gpu memory mb. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.58	1.86e-06	rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	—	—	constant: invalid
Llama 3.1 8B	MMLU-Pro	FP16	20	—	—	constant: invalid
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	—	—	constant: invalid
Llama 3.1 8B	RACE-H	FP16	20	—	—	constant: invalid
Llama 3.1 8B	RACE-H	Q4_K_M	20	—	—	constant: invalid
Qwen 2.5 7B	MMLU	FP16	20	0.433	8.32e-08	rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.351	1.85e-08	rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	—	—	constant: invalid
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	—	—	constant: invalid
Qwen 2.5 7B	RACE-H	FP16	20	—	—	constant: invalid
Qwen 2.5 7B	RACE-H	Q4_K_M	20	—	—	constant: invalid
Gemma 2 9B	MMLU	FP16	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.351	1.85e-08	rejected
Gemma 2 9B	MMLU-Pro	FP16	20	—	—	constant: invalid
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	—	—	constant: invalid
Gemma 2 9B	RACE-H	FP16	20	0.236	2.69e-09	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	—	—	constant: invalid

Table 34: Shapiro–Wilk checks: peak ram mb. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.884	0.021	rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.896	0.034	rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.906	0.053	not rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.784	0.000506	rejected
Llama 3.1 8B	RACE-H	FP16	20	0.928	0.143	not rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.88	0.018	rejected
Qwen 2.5 7B	MMLU	FP16	20	0.809	0.001	rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.866	0.01	rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.806	0.001	rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.916	0.084	not rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.917	0.085	not rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.88	0.018	rejected
Gemma 2 9B	MMLU	FP16	20	0.883	0.02	rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.97	0.752	not rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.835	0.003	rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.915	0.081	not rejected
Gemma 2 9B	RACE-H	FP16	20	0.964	0.636	not rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.907	0.056	not rejected

Table 35: Shapiro–Wilk checks: throughput tokens s. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.351	1.85e-08	rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.351	1.85e-08	rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.351	1.85e-08	rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.495	2.89e-07	rejected
Llama 3.1 8B	RACE-H	FP16	20	0.351	1.85e-08	rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.433	8.32e-08	rejected
Qwen 2.5 7B	MMLU	FP16	20	0.608	3.54e-06	rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.351	1.85e-08	rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.626	5.6e-06	rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.351	1.85e-08	rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.641	8.1e-06	rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU	FP16	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.58	1.86e-06	rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.236	2.69e-09	rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.626	5.6e-06	rejected
Gemma 2 9B	RACE-H	FP16	20	0.433	8.32e-08	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.495	2.89e-07	rejected

Table 36: Shapiro–Wilk checks: total energy j. Constant samples are invalid, not passed.

model_family	benchmark	precision	n	W	p_raw	status
Llama 3.1 8B	MMLU	FP16	20	0.949	0.349	not rejected
Llama 3.1 8B	MMLU	Q4_K_M	20	0.968	0.714	not rejected
Llama 3.1 8B	MMLU-Pro	FP16	20	0.949	0.346	not rejected
Llama 3.1 8B	MMLU-Pro	Q4_K_M	20	0.936	0.202	not rejected
Llama 3.1 8B	RACE-H	FP16	20	0.93	0.155	not rejected
Llama 3.1 8B	RACE-H	Q4_K_M	20	0.983	0.966	not rejected
Qwen 2.5 7B	MMLU	FP16	20	0.972	0.794	not rejected
Qwen 2.5 7B	MMLU	Q4_K_M	20	0.96	0.547	not rejected
Qwen 2.5 7B	MMLU-Pro	FP16	20	0.978	0.905	not rejected
Qwen 2.5 7B	MMLU-Pro	Q4_K_M	20	0.977	0.884	not rejected
Qwen 2.5 7B	RACE-H	FP16	20	0.941	0.247	not rejected
Qwen 2.5 7B	RACE-H	Q4_K_M	20	0.945	0.296	not rejected
Gemma 2 9B	MMLU	FP16	20	0.966	0.669	not rejected
Gemma 2 9B	MMLU	Q4_K_M	20	0.975	0.85	not rejected
Gemma 2 9B	MMLU-Pro	FP16	20	0.952	0.4	not rejected
Gemma 2 9B	MMLU-Pro	Q4_K_M	20	0.968	0.719	not rejected
Gemma 2 9B	RACE-H	FP16	20	0.345	1.65e-08	rejected
Gemma 2 9B	RACE-H	Q4_K_M	20	0.978	0.905	not rejected

Table 37: Levene checks: average power w; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	0.677	0.416	not rejected
Llama 3.1 8B	MMLU-Pro	6.937	0.012	rejected
Llama 3.1 8B	RACE-H	2.645	0.112	not rejected
Qwen 2.5 7B	MMLU	0.529	0.471	not rejected
Qwen 2.5 7B	MMLU-Pro	2.359	0.133	not rejected
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	1.743	0.195	not rejected
Gemma 2 9B	MMLU-Pro	12.038	0.001	rejected
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 38: Levene checks: cpu utilization pct; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	0.089	0.767	not rejected
Llama 3.1 8B	MMLU-Pro	1.358	0.251	not rejected
Llama 3.1 8B	RACE-H	—	—	not run: normality gate
Qwen 2.5 7B	MMLU	0.599	0.444	not rejected
Qwen 2.5 7B	MMLU-Pro	0.009	0.925	not rejected
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	2.191	0.147	not rejected
Gemma 2 9B	MMLU-Pro	0.7	0.408	not rejected
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 39: Levene checks: gpu utilization pct; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	—	—	not run: normality gate
Llama 3.1 8B	MMLU-Pro	—	—	not run: normality gate
Llama 3.1 8B	RACE-H	—	—	not run: normality gate
Qwen 2.5 7B	MMLU	—	—	not run: normality gate
Qwen 2.5 7B	MMLU-Pro	—	—	not run: normality gate
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	0.803	0.376	not rejected
Gemma 2 9B	MMLU-Pro	7.842	0.008	rejected
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 40: Levene checks: inference time s; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	—	—	not run: normality gate
Llama 3.1 8B	MMLU-Pro	—	—	not run: normality gate
Llama 3.1 8B	RACE-H	—	—	not run: normality gate
Qwen 2.5 7B	MMLU	—	—	not run: normality gate
Qwen 2.5 7B	MMLU-Pro	—	—	not run: normality gate
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	—	—	not run: normality gate
Gemma 2 9B	MMLU-Pro	—	—	not run: normality gate
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 41: Levene checks: peak gpu memory mb; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	—	—	not run: normality gate
Llama 3.1 8B	MMLU-Pro	—	—	not run: normality gate
Llama 3.1 8B	RACE-H	—	—	not run: normality gate
Qwen 2.5 7B	MMLU	—	—	not run: normality gate
Qwen 2.5 7B	MMLU-Pro	—	—	not run: normality gate
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	—	—	not run: normality gate
Gemma 2 9B	MMLU-Pro	—	—	not run: normality gate
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 42: Levene checks: peak ram mb; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	—	—	not run: normality gate
Llama 3.1 8B	MMLU-Pro	—	—	not run: normality gate
Llama 3.1 8B	RACE-H	—	—	not run: normality gate
Qwen 2.5 7B	MMLU	—	—	not run: normality gate
Qwen 2.5 7B	MMLU-Pro	—	—	not run: normality gate
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	—	—	not run: normality gate
Gemma 2 9B	MMLU-Pro	—	—	not run: normality gate
Gemma 2 9B	RACE-H	2.41	0.129	not rejected

Table 43: Levene checks: throughput tokens s; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	—	—	not run: normality gate
Llama 3.1 8B	MMLU-Pro	—	—	not run: normality gate
Llama 3.1 8B	RACE-H	—	—	not run: normality gate
Qwen 2.5 7B	MMLU	—	—	not run: normality gate
Qwen 2.5 7B	MMLU-Pro	—	—	not run: normality gate
Qwen 2.5 7B	RACE-H	—	—	not run: normality gate
Gemma 2 9B	MMLU	—	—	not run: normality gate
Gemma 2 9B	MMLU-Pro	—	—	not run: normality gate
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 44: Levene checks: total energy j; median centered, only when both normality checks do not reject.

model_family	benchmark	statistic	p_raw	status
Llama 3.1 8B	MMLU	2.355	0.133	not rejected
Llama 3.1 8B	MMLU-Pro	3.536	0.068	not rejected
Llama 3.1 8B	RACE-H	0.503	0.482	not rejected
Qwen 2.5 7B	MMLU	1.121	0.296	not rejected
Qwen 2.5 7B	MMLU-Pro	4.114	0.05	rejected
Qwen 2.5 7B	RACE-H	0.005	0.941	not rejected
Gemma 2 9B	MMLU	8.416	0.006	rejected
Gemma 2 9B	MMLU-Pro	0.822	0.37	not rejected
Gemma 2 9B	RACE-H	—	—	not run: normality gate

Table 45: average power w: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	256.512	236.25	-7.899	Student (d)	-2.645	3.8e-10	5.26e-10	True
Llama 3.1 8B	MMLU-Pro	314.401	354.466	12.743	Welch (d)	25.938	1.16e-38	3.47e-38	True
Llama 3.1 8B	RACE-H	283.464	281.078	-0.842	Student (d)	-0.25	0.434	0.434	False
Qwen 2.5 7B	MMLU	250.538	223.135	-10.938	Student (d)	-3.195	2.57e-12	3.85e-12	True
Qwen 2.5 7B	MMLU-Pro	331.08	354.017	6.928	Student (d)	20.627	1.27e-40	4.58e-40	True
Qwen 2.5 7B	RACE-H	287.478	275.911	-4.023	MWU (delta)	-0.755	4.68e-05	5.27e-05	True
Gemma 2 9B	MMLU	265.591	243.801	-8.205	Student (d)	-2.611	5.26e-10	6.76e-10	True
Gemma 2 9B	MMLU-Pro	321.062	338.755	5.511	Welch (d)	14.661	2.33e-25	3.82e-25	True
Gemma 2 9B	RACE-H	292.86	290.136	-0.93	MWU (delta)	-0.735	7.41e-05	7.84e-05	True

Table 46: cpu utilization pct: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	7.014	7.186	2.455	Student (d)	3.426	3.52e-13	4.75e-12	True
Llama 3.1 8B	MMLU-Pro	6.819	6.862	0.637	Student (d)	3.225	1.97e-12	2.12e-11	True
Llama 3.1 8B	RACE-H	7.083	7.137	0.763	MWU (delta)	0.615	0.000921	0.000956	True
Qwen 2.5 7B	MMLU	7.073	7.179	1.5	Student (d)	2.024	1.62e-07	1.78e-07	True
Qwen 2.5 7B	MMLU-Pro	6.821	6.858	0.547	Student (d)	2.431	2.96e-09	9.37e-09	True
Qwen 2.5 7B	RACE-H	7.067	7.14	1.029	MWU (delta)	0.45	0.015	0.015	True
Gemma 2 9B	MMLU	7.024	7.133	1.544	Student (d)	1.955	3.19e-07	3.45e-07	True
Gemma 2 9B	MMLU-Pro	6.823	6.85	0.404	Student (d)	2.028	1.55e-07	1.74e-07	True
Gemma 2 9B	RACE-H	7.074	7.097	0.317	MWU (delta)	0.465	0.012	0.013	True

Table 47: gpu utilization pct: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	62.603	37.16	-40.642	MWU (delta)	-1	6.76e-08	7.81e-08	True
Llama 3.1 8B	MMLU-Pro	94.713	87.198	-7.934	MWU (delta)	-1	6.8e-08	7.81e-08	True
Llama 3.1 8B	RACE-H	58.301	45.431	-22.075	MWU (delta)	-1	6.76e-08	7.81e-08	True
Qwen 2.5 7B	MMLU	50.345	31.912	-36.613	MWU (delta)	-1	6.65e-08	7.81e-08	True
Qwen 2.5 7B	MMLU-Pro	94.404	86.667	-8.196	MWU (delta)	-1	6.79e-08	7.81e-08	True
Qwen 2.5 7B	RACE-H	56.207	44.629	-20.598	MWU (delta)	-1	6.73e-08	7.81e-08	True
Gemma 2 9B	MMLU	60.543	41.479	-31.488	Student (d)	-53.029	3.83e-56	2.07e-54	True
Gemma 2 9B	MMLU-Pro	92.571	83.381	-9.927	Welch (d)	-200.568	3.23e-55	8.73e-54	True
Gemma 2 9B	RACE-H	62.497	52.763	-15.574	MWU (delta)	-0.9	1.19e-06	1.26e-06	True

Table 48: inference time s: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	19.1	9.1	-52.356	MWU (delta)	-1	2.42e-09	9.34e-09	True
Llama 3.1 8B	MMLU-Pro	616.9	222.8	-63.884	MWU (delta)	-1	4.28e-09	1.06e-08	True
Llama 3.1 8B	RACE-H	16.1	11.15	-30.745	MWU (delta)	-1	3.3e-09	9.37e-09	True
Qwen 2.5 7B	MMLU	12.65	7.9	-37.549	MWU (delta)	-1	7.17e-09	1.25e-08	True
Qwen 2.5 7B	MMLU-Pro	757.6	316.1	-58.276	MWU (delta)	-1	7.86e-09	1.29e-08	True
Qwen 2.5 7B	RACE-H	15.5	11.05	-28.71	MWU (delta)	-1	6.17e-09	1.15e-08	True
Gemma 2 9B	MMLU	18.05	10.7	-40.72	MWU (delta)	-1	4.54e-09	1.06e-08	True
Gemma 2 9B	MMLU-Pro	507.95	216.6	-57.358	MWU (delta)	-1	5.72e-09	1.14e-08	True
Gemma 2 9B	RACE-H	22.15	15.2	-31.377	MWU (delta)	-1	5.72e-09	1.14e-08	True

Table 49: peak gpu memory mb: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	16690.588	6825.188	-59.108	MWU (delta)	-1	3.1e-09	9.37e-09	True
Llama 3.1 8B	MMLU-Pro	16689.188	6823.188	-59.116	MWU (delta)	-1	4.68e-10	2.53e-09	True
Llama 3.1 8B	RACE-H	16697.188	6831.188	-59.088	MWU (delta)	-1	4.68e-10	2.53e-09	True
Qwen 2.5 7B	MMLU	15220.888	5946.988	-60.929	MWU (delta)	-1	3.3e-09	9.37e-09	True
Qwen 2.5 7B	MMLU-Pro	15219.188	5945.188	-60.936	MWU (delta)	-1	4.68e-10	2.53e-09	True
Qwen 2.5 7B	RACE-H	15223.188	5951.188	-60.907	MWU (delta)	-1	4.68e-10	2.53e-09	True
Gemma 2 9B	MMLU	21617.088	9520.987	-55.956	MWU (delta)	-1	1.68e-09	7.58e-09	True
Gemma 2 9B	MMLU-Pro	21617.188	9521.188	-55.955	MWU (delta)	-1	4.68e-10	2.53e-09	True
Gemma 2 9B	RACE-H	21618.287	9523.188	-55.948	MWU (delta)	-1	7.43e-10	3.65e-09	True

Table 50: peak ram mb: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	3607.479	2900.686	-19.592	MWU (delta)	-1	6.8e-08	7.81e-08	True
Llama 3.1 8B	MMLU-Pro	3623.837	2944.844	-18.737	MWU (delta)	-1	6.8e-08	7.81e-08	True
Llama 3.1 8B	RACE-H	3597.17	2909.556	-19.115	MWU (delta)	-1	6.8e-08	7.81e-08	True
Qwen 2.5 7B	MMLU	3682.448	2940.475	-20.149	MWU (delta)	-1	6.8e-08	7.81e-08	True
Qwen 2.5 7B	MMLU-Pro	3655.769	2919.063	-20.152	MWU (delta)	-1	6.8e-08	7.81e-08	True
Qwen 2.5 7B	RACE-H	3622.492	2905.395	-19.796	MWU (delta)	-1	6.8e-08	7.81e-08	True
Gemma 2 9B	MMLU	4474.383	3421.112	-23.54	MWU (delta)	-1	6.8e-08	7.81e-08	True
Gemma 2 9B	MMLU-Pro	4493.361	3446.955	-23.288	MWU (delta)	-1	6.8e-08	7.81e-08	True
Gemma 2 9B	RACE-H	4375.91	3402.89	-22.236	Student (d)	-10.978	2.16e-30	3.89e-29	True

Table 51: throughput tokens s: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	43.623	70.73	62.14	MWU (delta)	1	2.42e-09	9.34e-09	True
Llama 3.1 8B	MMLU-Pro	55.344	134.65	143.295	MWU (delta)	1	4.28e-09	1.06e-08	True
Llama 3.1 8B	RACE-H	31.066	44.886	44.486	MWU (delta)	1	3.3e-09	9.37e-09	True
Qwen 2.5 7B	MMLU	39.583	63.393	60.15	MWU (delta)	1	7.17e-09	1.25e-08	True
Qwen 2.5 7B	MMLU-Pro	58.515	140.965	140.904	MWU (delta)	1	7.86e-09	1.29e-08	True
Qwen 2.5 7B	RACE-H	32.292	45.265	40.176	MWU (delta)	1	6.17e-09	1.15e-08	True
Gemma 2 9B	MMLU	32.969	55.339	67.854	MWU (delta)	1	4.54e-09	1.06e-08	True
Gemma 2 9B	MMLU-Pro	42.823	96.644	125.682	MWU (delta)	1	5.72e-09	1.14e-08	True
Gemma 2 9B	RACE-H	27.095	39.5	45.784	MWU (delta)	1	5.72e-09	1.14e-08	True

Table 52: total energy j: independent two-sided comparisons; effects use Q4 minus FP16.

Model	Benchmark	FP16 mean	Q4 mean	Change (%)	Test (effect)	Effect	Raw p	BH p	Reject
Llama 3.1 8B	MMLU	5806.049	2960.564	-49.009	Student (d)	-22.483	4.94e-42	2.22e-41	True
Llama 3.1 8B	MMLU-Pro	195011.516	80123.76	-58.913	Student (d)	-196.192	1.01e-77	9.05e-77	True
Llama 3.1 8B	RACE-H	5489.345	4131.25	-24.741	Student (d)	-8.314	5.52e-26	9.93e-26	True
Qwen 2.5 7B	MMLU	4050.505	2529.102	-37.561	Student (d)	-13.701	5.85e-34	1.5e-33	True
Qwen 2.5 7B	MMLU-Pro	251885.768	113081.45	-55.106	Welch (d)	-238.599	3.04e-68	1.82e-67	True
Qwen 2.5 7B	RACE-H	5444.672	3978.029	-26.937	Student (d)	-12.898	5.53e-33	1.24e-32	True
Gemma 2 9B	MMLU	5739.637	3437.64	-40.107	Welch (d)	-14.603	2.81e-28	5.61e-28	True
Gemma 2 9B	MMLU-Pro	164100.739	74496.372	-54.603	Student (d)	-265.838	9.75e-83	1.76e-81	True
Gemma 2 9B	RACE-H	7490.394	5347.205	-28.612	MWU (delta)	-0.9	1.2e-06	1.44e-06	True

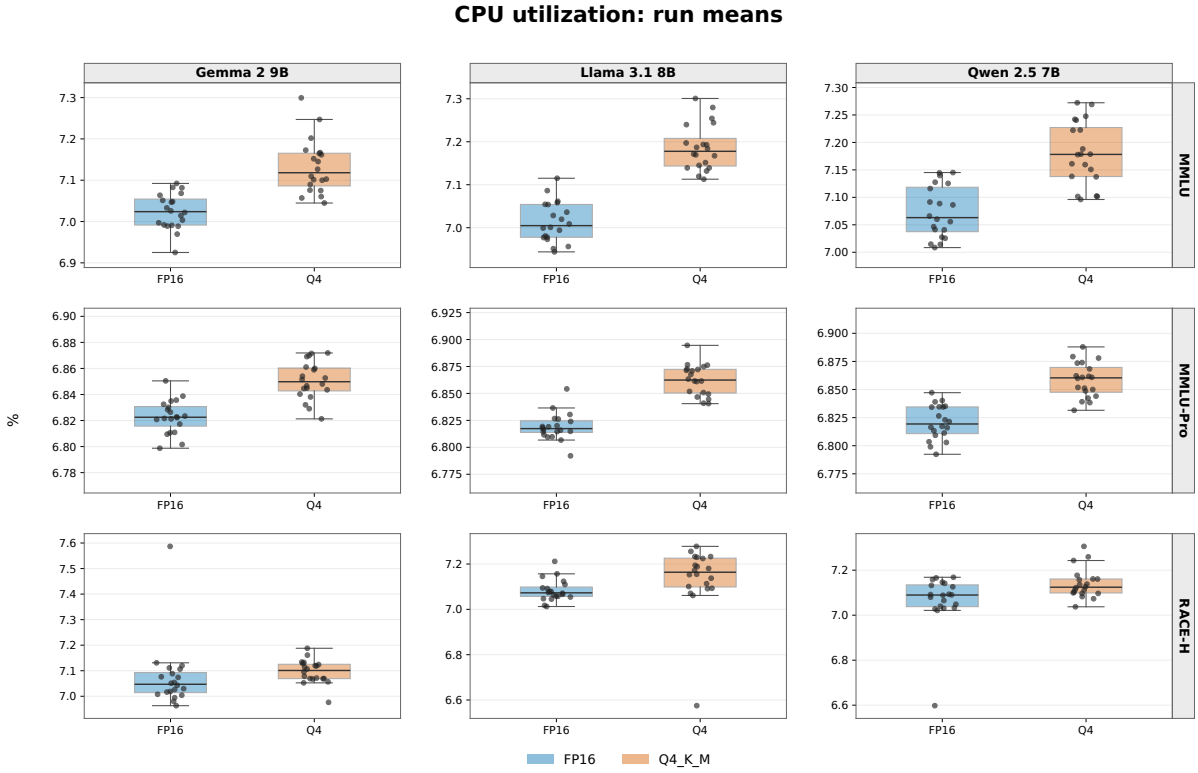


Figure 7: Distribution of run-level mean CPU utilization across 20 repetitions per configuration.

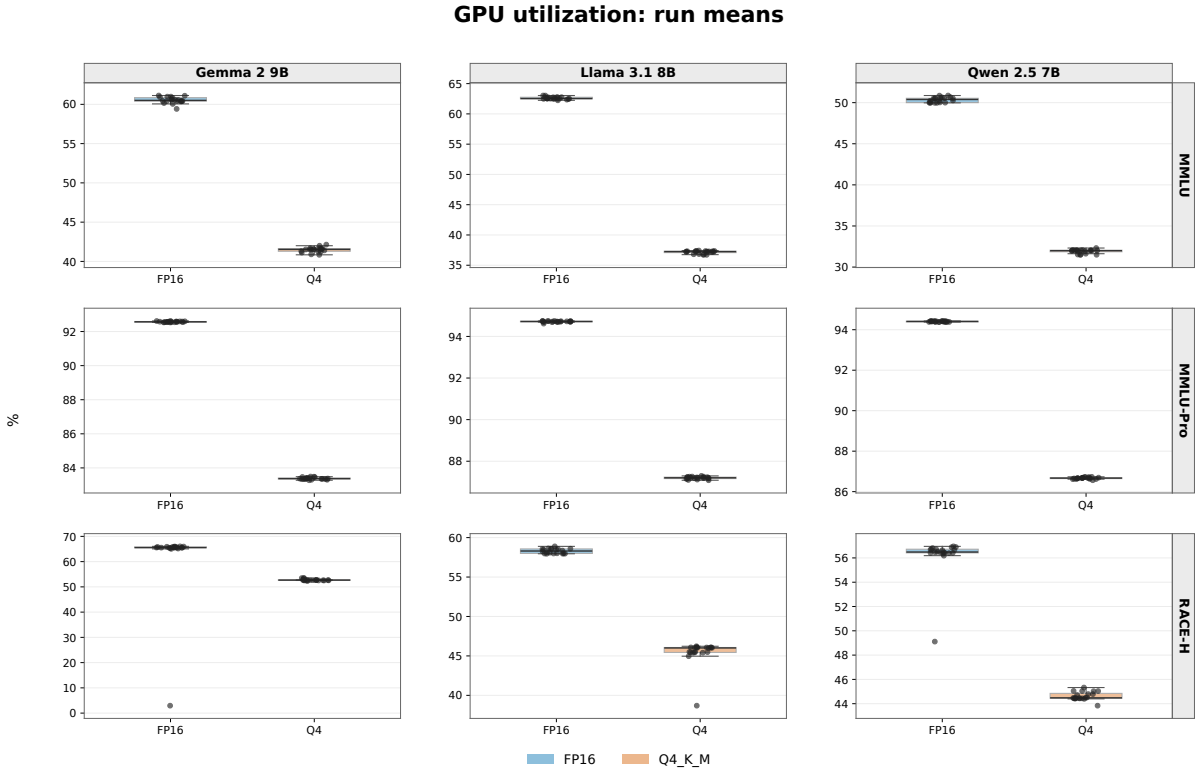


Figure 8: Distribution of run-level mean GPU utilization across 20 repetitions per configuration.